Eye

For concentrated complexities, no other organ in Joe's body can equal me. No larger than a Ping-Pong ball, I have tens of millions of electrical connections and can handle 1.5 million simultaneous messages. I gather 80 percent of all the knowledge Joe absorbs. He thinks of me as a miniature television camera. I consider the comparison insulting. I am much more sensitive than the biggest, costliest TV camera ever made. I am responsible for one of the greatest of all miracles - sight.

Today's world is giving me a hard time. I was not built for it. For Joe's prehistoric ancestors, the eye's main job was to see things at a distance danger to be avoided, game to be killed. Only lately have I been called on for continuous close-up work.

Look at my anatomy and you will understand why I am having difficulty adjusting to today's demands. First, my front window - my clear, 20 cent size cornea. It starts the seeing process by bending light rays into orderly patterns. Next, my pupil - an adjustable gateway for light. In bright sun it is nearly closed; on a dark night it is wide open. Up to this point there is nothing about seeing that a cheap camera could not handle.

My wonders really begin with my lens-a little envelope of fluid the size and shape of an oval vitamin pill. My lens is surrounded by a ring of tiny, superbly strong, unbelievably hard-working muscles. When they tense, my lens fattens for near vision; when they relax, it flattens for distant vision. This was a fine arrangement for Joe's caveman ancestors. Since they were mainly interested in things six or more meters away, the muscles were relaxed most of the time. But Joe now lives in a close-up world-lots of reading, deskwork and such. This keeps my ciliary muscles tensed a lot of the time. They grow tired.

In front of and behind my lens I have two fluid-filled chambers In front the fluid is like water; in back it is about the consistency of egg white. The watery fluid keeps me firmly inflated. Both fluids must be absolutely clear to permit passage of light. Those "specks" Joe sees when he looks at a bright light are cellular remnants left over from his days in the womb when I was under construction. They will float aimlessly in his eye fluid as long as he lives.

When Joe looks at some object, the light passes through my lens, which brings it into correct focus on my retina, a kind of onionskin wallpaper, which covers the rear two thirds of my interior. Except in Joe's brain, I do not think that anywhere else in his body is so much packed into so small a space. Covering less than three-square centimeters, my retina contains 137 million light-sensitive receptor cells 130 million shaped like rods for black-and-white vision, seven million shaped like cones for color vision.

The rods are scattered all over my retina. Let a firefly pass at night and a complex chemistry gets under way. The faint light bleaches rhodopsin, a purplish-red pigment in my rods. The bleaching process generates a tiny wisp of electricity--a few millionths of a volt, far

too little to tickle a mosquito. This feeds into my straw-size optic nerve and is transmitted to Joe's brain at about 450 km. per hour. The brain interprets the signals flooding in and hands down its verdict: a firefly. All of this intricate electrochemical activity has been completed in about .002 second!

If my rods seem complex, my cones are far more so. They are concentrated in the fovea, a pinhead size, and yellowish depression at the rear of my chamber. This is the center for acute vision - reading, any close work--and for color. A leading theory is that these cones, too, have bleachable pigments, one each for red, green and blue. Like an artist mixing paints on a palette, Joe's brain blends these colors to make scores of other hues. If anything should go wrong with this intricate electrochemical process, Joe would be - color-blind--as one in eight men is to some degree. In dim light, activity of my cones diminishes, color sense vanishes and every thing becomes gray, as my rods take over.

While Joe sees with me, he sees in his brain. A crushing blow at the back of his head, severe enough to destroy the optical center of his brain, would produce permanent blindness. A lesser blow and he sees “stars” - a chaotic electrical disturbance. Joe gets clinching evidence of the brain's role when he dreams. He "sees" pictures, even with my lids closed in total darkness. Had he been born blind, he would dream in terms of other sensory stimuli: touch, sound, even smell.

Joe was not born with the eyes he has today. At birth he could see only light and shadow. In the first few months he was farsighted, like his caveman ancestors. To study his rattle, he held it as far as possible from his face. At first, Joe's eyes were poorly coordinated; I would wander in one direction, my partner in another. Our wandering worried Joe's mother. It should not have. A few months after birth we were moving in exact unison. By the time Joe was six, his vision was excellent. But my peak sight did not come until age eight.

When he was young, Joe used to read in dim light. His mother warned that he was "ruining" his eyes. Nonsense. The young see better in dim light than adults; and viewing under even the most adverse circumstances does no harm.

I have a number of other unusual attributes. Tiny though they are, my muscles, milligram for milligram, are among the body's strongest. In an average day, I move about 100,000 times - to bring objects into sharp focus. Joe would have to walk 80km to give his leg muscles similar exercise.

My cleaning equipment is similarly striking. My lacrimal glands produce a steady stream of moisture--tears to flush away dust and other foreign material. My eyelids, of course, act as windshield wipers. Joe blinks three to six times a minute- more when I am tired. This keeps my cornea moist and clean. The tears also contain a potent microbe-killer called lysozyme, which guards me from infective bacteria.

I try to ward off fatigue by resting as much as possible. I get time oh when Joe blinks. And my partner and I spell each other for a while I may carry 90 percent of the work load, while Joe's other eye loafs; then it goes to work and I rest.

Nature gave me superb protection, placing me in a bony cavern with protruding cheekbones and forehead to act as shock absorbers for direct blows. She also gave me supersensitive nerves to activate the alarm if there should be a potentially damaging intruder such as cinder.

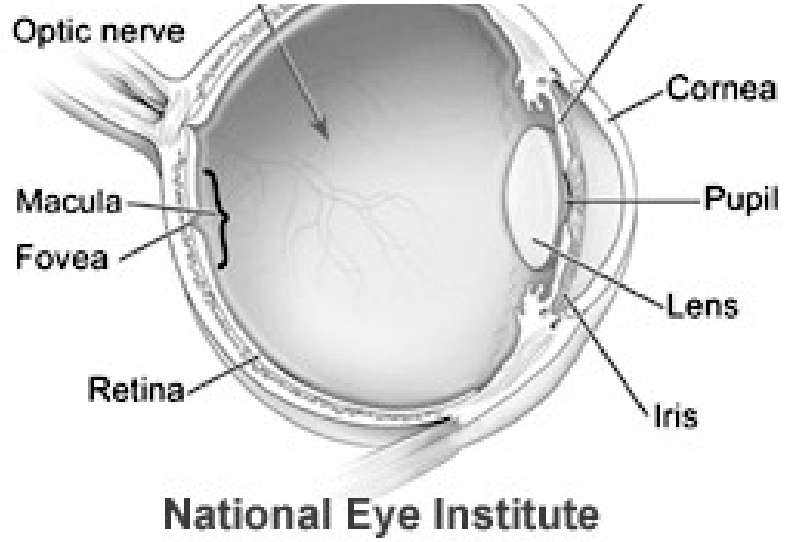
Still, I do have my troubles. My focusing apparatus often fails to work perfectly. Eyeglasses can correct 95 percent of this trouble. Disease is a more serious problem. One potential disorder is really a plumbing problem either too much fluid entering me or too little draining away. Pressure builds up, reducing the blood supply to my optic nerve. This is
glaucoma.

In severe instances, glaucoma can cause permanent blindness in a few days. More often it is a leisurely performer, producing symptoms so mild they are apt to go unnoticed. These symptoms: colored halos around bright lights, loss of side vision, difficulty in adjusting to the dark, a blurring of vision. At his age, Joe has one chance in 40 of glaucoma damaging his sight or blinding him completely. His doctor can check for glaucoma simply by pressing a little gadget called a tonometer against my eyeball. Joe should have this test every year. The treatment for glaucoma? Drugs in drop
form, or surgery.

Astigmatism is another of my common ailments. In this one, my cornea is not a spherical surface and distorts vision like a bubble flaw in a piece of glass. Eyeglasses correct this condition A detached retina is more serious. It occurs when my retinal wallpaper blisters or peels, and usually announces itself with flashing lights, image distortion blurring spots. A surgeon can "tack" my wallpaper back in place with 80 percent chance of success.

Both my cornea and my lens-normally totally transparent tissue--can cloud and lead to blindness. If it is the cornea, Joe can re gain sight with a corneal transplant. If it is the lens, he will need a cataract operation, and either thick eyeglasses or contact lenses afterward.

Fortunately, Joe has so far escaped all these things. Just the same, I am growing old--like Joe's other organs. The transparency of my lens is lower, accommodation muscles are weaker, hardened arteries are diminishing blood supply to my retina. These processes will continue, but Joe should not worry unduly. The odds are heavily in favor of my providing him with serviceable vision as long as he lives.



Questions

1. What is the role of the iris? (Use the term dilate and constrict in your answer)
2. What is the role of the lens?
3. What is the role of the retina?
4. What is the role of rods and cones in the eye?
5. Compare myopia with hyperopia.
6. What causes glaucoma?
7. Why do we blink?